AIGR Gene Hypothesis Deep Research — AMD2 (P22580) Amidase Activity (GO:0004040)

Target gene: AMD2 / YDR242W, *Saccharomyces cerevisiae* (NCBITaxon:559292) **UniProt:** P22580 · **Hypothesis:** *AMD2 has amidase activity (GO:0004040)* **Focus type:** function_assignment · **Evidence type on record:** IEA (GO_REF:0000120) **Source:** genes/yeast/AMD2/AMD2-ai-review.yaml → existing_annotations[1].function_hypothesis

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Summary

The hypothesis that AMD2 has amidase activity (GO:0004040) is **supported at the enzyme-family level but is entirely homology-inferred**, and the recommended curation action is to **retain GO:0004040 at its current general level — neither narrow it to a specific substrate nor remove it**. AMD2/YDR242W (UniProt P22580, 549 aa, annotated "Probable amidase") is an unambiguous member of the **amidase-signature (AS) superfamily**: it matches Pfam PF01425, PROSITE PS00571, and InterPro IPR023631/IPR020556/IPR036928, and is additionally classified by the fungal-specific PIRSF001221 "Amidase, fungal type." The diagnostic AS catalytic machinery is intact — the signature motif GGSSGGE sits at residues 207–213 and the canonical **Lys–cisSer–Ser catalytic triad (K176/S210/S238)** is present and, per the AlphaFold model (AF-P22580-F1, global pLDDT 96.25), confidently and well-ordered in three dimensions. UniProt records the family reaction *monocarboxylic acid amide* + H_2O → *monocarboxylate* + NH_4^+, which is exactly the reaction defining GO:0004040.

The crucial caveat is that **no experimental evidence exists** for AMD2's activity. UniProt lists protein existence at level 3 ("Inferred from homology"), SGD describes the product as a "Putative amidase," and the founding reference (P 2263500) is a sequence-identification-by-homology study rather than a biochemical characterization. Because the AS superfamily is functionally heterogeneous — spanning fatty-acid amide hydrolase, GatA glutamyl-tRNA amidotransferase, indoleacetamide hydrolase, acetamidase/

formamidase, malonamidase, and nicotinamidase-like activities — family membership alone cannot fix a substrate. A global alignment against characterized AS enzymes shows the closest characterized homolog is the fungal acetamidase **AmdS** (*Aspergillus nidulans*, P08158, EC 3.5.1.4) at only **33.2% identity**, well below the ~40–50% typically required to transfer a specific EC/substrate.

The net picture is a structurally competent, fungal-type amidase of **undetermined substrate**. GO:0004040 (amidase activity) sits at precisely the right level of the ontology for this evidence: it asserts the amide-hydrolase chemistry without committing to a specific substrate. Narrowing to acetamidase (GO:0004039) would be over-annotation; removing the term would ignore strong, mutually consistent domain and structural evidence. The IEA evidence code honestly reflects the homology-only basis and should be retained, ideally with a curator note flagging the absence of experimental substrate data.

Key Findings

F001 — AMD2 carries an intact amidase-signature domain with a conserved catalytic triad

AMD2/YDR242W is a canonical member of the amidase-signature (AS) superfamily, and this is the strongest single line of support for the hypothesis. UniProt P22580 (549 aa, "Probable amidase") matches **Pfam PF01425 (Amidase)**, **PROSITE PS00571 (AMIDASES, MatchStatus = 1)**, and the InterPro entries **IPR023631 (Amidase signature domain)**, **IPR020556 (Amidase conserved site)**, and **IPR036928 (AS superfamily)**, along with the structural classifiers Gene3D 3.90.1300.10 and SUPFAM SSF75304. Direct analysis of the primary sequence locates the diagnostic AS signature motif **GGSSGGE at residues 207–213** (catalytic nucleophile Ser ≈ S210) and the catalytic Lys within the F(I)VKTT motif (≈ K176), completing the AS-family **Ser–cisSer–Lys catalytic triad** that defines this enzyme class' mechanism. Critically, the InterPro "Amidase, conserved site" **catalytic-site profile** matches, which means the catalytic residues are present and intact — this is an active-enzyme signature, not a decayed pseudoenzyme relic. UniProt records the catalytic activity as **monocarboxylic acid amide** + **H_2O** = **monocarboxylate** + **NH_4^+** (an EC 3.5.1.4-type reaction), the exact reaction that GO:0004040 describes. Taken together, every sequence-level prerequisite for amidase chemistry is satisfied.

F002 — The amidase activity is homology-inferred only; no experimental substrate characterization exists

Despite the strong domain evidence, no direct biochemical evidence supports a *specific* enzymatic activity for AMD2. UniProt lists **protein existence = "3: Inferred from homology"**; the keyword set is limited to "Hydrolase"; and the only comment beyond the (inferred) catalytic-activity line is a SIMILARITY note ("Belongs to the amidase family"). SGD (YDR242W) classifies the ORF as **"Verified"** — meaning it is a real, expressed gene — but describes its product only as a **"Putative amidase."** The founding reference (P 2263500) is a sequence-identification-by-homology paper, not an enzyme characterization. Because the AS superfamily is functionally heterogeneous, membership does not determine a substrate. Supporting context comes from work on the bottom-fermenting-yeast amidase AMI1 (PMID: 17924455), which reports that the canonical fungal amidase AMI1 is **conserved among plants, *Bacillus subtilis*, *Neurospora crassa*, *Schizosaccharomyces pombe* and *Saccharomyces* species, "with the exception of *S. cerevisiae* S288C."** This establishes that AMD2 is a *distinct* amidase-signature gene — not the canonical fungal AMI1 amidase — and that yeast AS enzymes function as **general amide hydrolases** ("suggesting that Ami1p may hydrolyse some amides related to amino acid and niacin metabolism in the cell"). Both observations reinforce a **family-level GO:0004040 assignment** rather than a narrow, substrate-specific term.

F003 — The AlphaFold model is high-confidence with a well-ordered catalytic triad (K176–S210–S238)

The AlphaFold model AF-P22580-F1 has a **global mean pLDDT of 96.25** (median per-residue 97.7 over 549 residues), placing it firmly in the very-high-confidence regime. The amidase-signature catalytic residues are all confidently modeled: **Lys176 pLDDT = 98.9, Ser210 pLDDT = 98.7, Ser238 pLDDT = 98.8**. The InterPro/PROSITE conserved catalytic site spans residues 207–238 and contains both catalytic serines of the AS Lys–cisSer–Ser triad, structurally analogous to the well-characterized human FAAH triad (Lys142/Ser217/Ser241). Domain boundaries are consistent across member databases — Pfam PF01425 (79–539), Gene3D AS domain (17–546), SUPFAM SSF75304 (26–545) — and, importantly, **PIRSF001221 "Amidase, fungal type"** spans the full protein (3–549). PIRSF is a fungal-amidase-specific classifier that goes beyond the generic AS superfamily assignment, reinforcing that AMD2 is a fungal-type amidase with a structurally competent active site. In short, this is a protein built to catalyze amide hydrolysis; the model shows no active-site occlusion or triad disruption. This is structural *potential*, not demonstrated activity, but it removes the possibility that AMD2 is a degenerate pseudoenzyme.

F004 — The closest characterized homolog is fungal acetamidase AmdS (33% identity) — supports general amidase, not a specific substrate

To test whether the family-level annotation could be legitimately refined, a Needleman–Wunsch global alignment (match +2 / mismatch −1 / gap −2) was run between AMD2 (549 aa) and a panel of *characterized* AS enzymes:

Reference enzyme	UniProt	EC	Length	% identity to AMD2
A. nidulans AmdS acetamidase	P08158	3.5.1.4	548 aa	**33.2%** (closest)
Human FAAH (fatty-acid amide hydrolase)	O00519	3.5.1.99	—	27.6%
Agrobacterium indoleacetamide hydrolase	P06618	3.5.1.—	—	24.9%

(Three additional candidate accessions resolved to non-AS proteins — kynurenine 3-monooxygenase, snRNP-F, and an unannotated fragment — and were excluded as accession mismatches.) AMD2 is therefore most similar to a **fungal aliphatic amidase/acetamidase** acting in nitrogen-source amide utilization, consistent with its PIRSF001221 "Amidase, fungal type" classification. However, **33% identity is well below the ~40–50% threshold** typically required to confidently transfer a specific substrate or EC number. The comparison usefully narrows the likely functional neighborhood (a fungal amide hydrolase probably tied to nitrogen/amide metabolism) without licensing a specific substrate call — precisely the situation in which GO best practice is to annotate the general parent term (GO:0004040) rather than a specific child.

Mechanistic Model / Interpretation

The four findings converge on a single, coherent picture: **AMD2 is a structurally competent, fungal-type amidase-signature enzyme of undefined substrate.**

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Sequence              Domain / motif            Structure (AlphaFold)       Nearest characteri
(P22580, 549 aa)      evidence                  pLDDT 96.25                 homolog
───────────────       ─────────────────────     ─────────────────────     ─────────────────
"Probable             Pfam PF01425              Catalytic triad             AmdS acetamidase
 amidase"             PROSITE PS00571           K176 (98.9)                 (P08158, EC 3.5.1.4
                      IPR023631 (AS domain)     S210 (98.7)  ── well        33.2% identity
Signature motif       IPR020556 (cat. site)     S238 (98.8)     ordered     ──────────────
GGSSGGE (207-213)     PIRSF001221 (fungal)      Active site intact          FAAH  27.6%
                                                                            IaaH  24.9%
     │                     │                          │                           │
     └─────────────────────┴──────────────┬───────────┴───────────────────────────┘
                                          ▼
                  Confident enzyme-FAMILY assignment: amidase activity
                  (GO:0004040, EC 3.5.1.- : R-CO-NH2 + H2O -> R-COO- + NH4+)
                                          │
                                          ▼
                  BUT: no assay, no measured substrate, no kinetics
                  (UniProt PE level 3; SGD "Putative amidase")
                                          │
                                          ▼
                  Substrate/EC UNDETERMINED  →  keep GO:0004040 GENERAL
                  (do not narrow to acetamidase; do not remove)
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The immediate molecular function being tested is **hydrolysis of a carbon–nitrogen amide bond**: R–CO–NH_2 + H_2O → R–COO^- + NH_4^+, catalyzed by the AS-family Ser–cisSer–Lys triad in which the nucleophilic serine attacks the amide carbonyl. Every structural prerequisite for this chemistry is present and confidently modeled in AMD2. What is missing is the *identity of R* — the physiological substrate. GO:0004040 (amidase activity) sits at exactly the right level of the ontology for what the evidence supports: it asserts amide-hydrolase chemistry without committing to a specific substrate (e.g., acetamide → GO:0004039, or fatty-acid amide → GO:0017064). Given 33% identity to the nearest characterized enzyme, committing to any child term would over-reach.

Importantly, there is no evidence that the annotation reflects a downstream phenotype, a pleiotropic effect, or a loss-of-function inference — it is a direct molecular-function prediction grounded in an intact catalytic domain. The weakness is not that the wrong *type* of function was inferred; it is that the inference has not been experimentally validated and cannot yet be refined to a substrate.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confide limitatio
UniProt P22580 (record)	Review/ database, sequence-domain	**Supports**	AMD2 is an amidase	549 aa "Probable amidase"; Pfam PF01425, PROSITE PS00571; catalytic activity = monocarboxylic amide + H_2O → monocarboxylate + NH_4^+	*S. cerevisiae* record, in silico	High for member level 3, n protein evidence
InterPro IPR023631/ IPR020556/ IPR036928; PIRSF001221	Structural/ evolutionary (computational)	**Supports**	Intact AS catalytic machinery	AS domain + "conserved site" catalytic-site profile match; fungal-type classifier spans 3–549	Domain databases	High; mo match ≠ demons turnover
AlphaFold AF-P22580-F1 (own analysis)	Structural (computational)	**Supports**	Active site is structurally competent	Global pLDDT 96.25; K176/S210/ S238 at pLDDT 98.7–98.9; canonical AS fold, ordered triad	Predicted structure	High mo confiden predictio experim structur activity
Global alignment vs characterized AS enzymes (own analysis)	Comparative/ evolutionary (computational)	**Qualifies**	Can the term be narrowed?	Closest characterized homolog = AmdS acetamidase (EC 3.5.1.4) at 33.2%; FAAH 27.6%; IaaH 24.9%	Cross-species AS enzymes	33% too transfer EC; narr neighbo only
PMID: 17924455	Primary / comparative	**Qualifies**	Is AMD2 the canonical	Canonical AMI1 amidase absent from *S. cerevisiae*	Bottom-fermenting	AMD2 is distinct gene; su

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confide limitatic
	genomics + overexpression		fungal amidase?	S288C; yeast AS enzymes act as general amide hydrolases (amino-acid/ niacin amides)	yeast (S. pastorianus)	general specific)
PMID: 32024536	Primary / functional genetics	**Qualifies**	Do yeasts have functional acetamidases?	*Y. lipolytica* acetamidase (AmdS homolog) confers acetamide utilization; usable as marker in *S. cerevisiae*	*Yarrowia lipolytica*	Establisl plausibl acetami type fun yeast, bu other ge AMD2
PMID: 9126617	Primary / regulatory genetics	Context	amdS/ amidase regulation in fungi	Characterizes amdA regulation of the amdS acetamidase system	*A. nidulans*	Orientat AMD2's characte homolog pathway
P 2263500 (founding reference)	Primary / sequence	**Supports (weakly)**	Gene identity	Original identification of AMD2 as a putative amidase by homology	*S. cerevisiae*	Homolog not a biochem characte
SGD YDR242W (record)	Database	Supports/ Qualifies	ORF is real; function putative	Status "Verified" ORF; description "Putative amidase"	*S. cerevisiae*	High tha real; fur unprove

Key literature notes:

- **PMID: 17924455** — *Identification and characterization of amidase-homologous AMI1 genes of bottom-fermenting yeast.* Establishes that the canonical fungal amidase AMI1 is present across fungi/plants/bacteria **but absent from *S. cerevisiae* S288C**, and that yeast AS amidases behave as general amide hydrolases (amino-acid/niacin amides). Directly supports treating AMD2 as a *distinct*, general amidase-signature enzyme rather than a substrate-specific one.

- **PMID: 32024536** — *Identification of a Yarrowia lipolytica acetamidase and its use as a yeast genetic marker.* Shows a functional yeast acetamidase (AmdS homolog) that works as a marker in *S. cerevisiae*, supporting the plausibility of acetamidase-type function in yeasts — while highlighting that the activity was demonstrated in *Y. lipolytica*, not AMD2, reinforcing "general term only."
- **PMID: 9126617** — *The amdA regulatory gene of Aspergillus nidulans.* Orientation on the amdS/acetamidase regulatory system, the pathway of AMD2's nearest characterized homolog.

(P 17110954 (Hst2/Sir2) and P 12095683 (Xenopus Polycomb) in the literature pool concern unrelated proteins and are not relevant to AMD2; noted only for completeness.)

GO Curation Implications

Lead (requires curator verification): *Retain GO:0004040 (amidase activity), Molecular Function, as a general homology-based (IEA) annotation; do not narrow, do not remove.*

- **Ontology aspect:** This is a **Molecular Function (MF)** assertion. The evidence (intact AS domain, conserved catalytic-site profile, high-confidence triad) supports the MF term at the level of "amidase activity." No Biological Process (BP) or Cellular Component (CC) term is supported by the current evidence for AMD2 specifically — substrate, pathway, and localization are all undetermined. (A general companion BP term such as amide catabolic process, if present as IBA, is consistent and may remain, but is likewise not substrate-specific.)
- **Retain vs generalize vs specify: Retain at the current general level.** The term should *not* be made more specific (e.g., acetamidase activity GO:0004039) because the nearest characterized homolog is only 33% identical — below the confidence threshold for substrate transfer. It should *not* be removed, because the domain and structural evidence

for amide-hydrolase chemistry is strong and consistent, and the ORF is "Verified" (no pseudogene signal).

- **Evidence code:** The IEA code (GO_REF:0000120) is **appropriate** and honestly reflects the homology-only basis. No experimental evidence warrants an upgrade to IDA/IMP.
- **Curator note:** Consider flagging that AMD2 is a *fungal-type* AS amidase (PIRSF001221) of **undetermined substrate**, distinct from the canonical fungal AMI1 amidase (absent from S288C), to prevent future over-annotation.

GO decision table

Candidate term	ID	Aspect	Recommendation	Rationale
amidase activity	GO:0004040	MF	**Retain (general, IEA)**	Intact AS domain + triad; correct level for evidence
acetamidase activity	GO:0004039	MF	**Do not assign**	Only 33% id to AmdS; substrate unproven
fatty acid amide hydrolase activity	GO:0017064	MF	Do not assign	27.6% id to FAAH; no evidence
nitrogen/amide catabolic process	(BP)	BP	Not supported for AMD2	No pathway data for this gene
any CC term	—	CC	Not supported	No localization data for AMD2

"Protein binding" avoidance: not applicable — a genuinely informative MF term (amidase activity) is supported, so no fallback to protein binding is needed or appropriate.

Mechanistic Scope

Directly tested (gene-product activity): the capacity of AMD2 to hydrolyze a carbon–nitrogen amide bond via an AS-family Ser–cisSer–Lys triad. The evidence for the *chemistry being possible* is strong (intact motif GGSSGGE at 207–213; confidently modeled K176/S210/S238; matching catalytic-site profile). The evidence for the *chemistry actually occurring on a defined substrate in vivo* is absent.

Explicitly NOT established (and not part of the supported annotation): - **Substrate identity** — no measured substrate; the AS family is substrate-diverse. - **Biological process** — no pathway placement (e.g., nitrogen-source utilization) demonstrated for AMD2. - **Localization** — no CC evidence for the AMD2 product. - **Phenotype** — no *amd2Δ* growth or metabolic phenotype cited; the annotation is not inferred from loss of function.

The hypothesis is therefore a **clean molecular-function assertion** whose only deficiency is validation depth (homology vs assay), not a case of a downstream phenotype masquerading as a direct function.

Conflicts and Alternatives

1. **Substrate ambiguity within a heterogeneous superfamily.** The main alternative to "acetamidase-like" is simply "some other amide hydrolase." Because the AS superfamily encompasses FAAH, GatA, indoleacetamide hydrolase, malonamidase, and nicotinamidase-like activities, the correct resolution of the seed hypothesis is the *general* term. This is a scope consideration, not a contradiction — it argues against narrowing, not against the term.

2. **AMD2 is not the canonical fungal AMI1 amidase.** PMID: 17924455 shows AMI1 is conserved across fungi/plants/bacteria *except* in *S. cerevisiae* S288C. AMD2 is thus a distinct AS gene, and one should not import AMI1's inferred substrate preferences (amino-acid/niacin amides) directly onto AMD2 — although they usefully illustrate the "general amide hydrolase" behavior of yeast AS enzymes.

3. **Acetamidase-type function is plausible but tested only in other species.** PMID: 32024536 demonstrates a functional *Y. lipolytica* acetamidase (AmdS homolog) that even works as a selectable marker in *S. cerevisiae* — implying *S. cerevisiae* itself lacks a strong native acetamidase for that selection. This is a mild *conflict* with narrowing AMD2 to acetamidase and reinforces "general term only."

4. **AMD2 ≠ AMD1.** AMD1 is AMP deaminase (unrelated); the shared "AMD" symbol is not a functional link and creates no paralog confusion in the GO term itself.

5. **Database carry-over risk.** The annotation traces to a homology-mapping GO_REF and a sequence-only founding reference. Any *specific* substrate label attached downstream would risk unsupported carry-over. The conservative retain-general action minimizes this risk.

No isoform-specific or experimental-artifact conflicts were identified that would undermine the *general* amidase call.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No enzymatic assay	UniProt PE level 3; SGD "Putative"; founding ref sequence-only	Distinguishes IEA-general from experimental (IDA) support	Purify recombinant AMD2; test amide-hydrolase activity on a substrate panel
Substrate unknown	AS family heterogeneous; nearest homolog only 33% id	Determines whether GO:0004040 can be narrowed to a child term	Substrate profiling (acetamide, fatty-acid amides, malonamide, aromatic amides) + kinetics
No demonstrated turnover despite intact residues	Motif/triad conservation established	Conservation ≠ activity; guards against pseudoenzyme	Catalytic-Ser mutant (S210A) vs WT activity comparison
No BP/pathway data	No pathway evidence for AMD2 specifically	Whether any BP term can be added	*amd2Δ* growth screens on amide nitrogen sources; metabolomics
No CC/localization data	No localization evidence found	Whether a CC term is warranted	GFP-tagging / high-throughput localization datasets
Analyses are computational	AlphaFold + global alignment run locally	Structure/alignment predict capability, not activity	Experimental structure with bound ligand; activity assay
GO_REF provenance	UniProt GO xref; record cites GO_REF:0000120	Affects how the IEA is attributed	Confirm against current UniProt-GOA source (vs EC2GO GO_REF:0000003)

This review is based on public sequence, domain, orthology, and database records plus literature; no local bioinformatics files were provided. Motif/triad conservation is a strong but indirect indicator — it establishes catalytic *potential*, not demonstrated activity or substrate. All computational results above are reported conservatively and distinguished from inference.

Discriminating Tests

The following would most efficiently distinguish "general amidase (undetermined substrate)" from specific alternatives, in rough priority order:

1. **In vitro amidase assay on recombinant AMD2 with a substrate panel.** Express and purify AMD2; assay against acetamide, formamide, malonamide, fatty-acid amides (e.g., oleamide), indoleacetamide, and nicotinamide, measuring NH_4^+ release or product formation. A positive hit on one substrate class would license a specific child term (e.g., GO:0004039 for acetamide) and upgrade the evidence code to IDA.
2. **Growth phenotyping of *amd2Δ* on amide nitrogen sources.** Test whether deletion impairs utilization of acetamide/formamide/other amides as sole nitrogen source — a BP-relevant readout that could place AMD2 in nitrogen metabolism (amdS-style assay, cf. PMID: 32024536).
3. **Site-directed mutagenesis of the triad (S210A / K176A).** Confirms catalytic dependence on the predicted triad and rules out a pseudoenzyme/moonlighting role.
4. **Structure with a bound substrate/transition-state analog.** Would define the substrate-binding pocket and predict specificity directly.
5. **Comparative substrate-specificity-determinant analysis** across the yeast AS paralog set and characterized homologs (AmdS, FAAH, malonamidase E2), mapping active-site residues that discriminate substrates.

Proposed Follow-up Experiments / Actions

For the curator (immediate, low-cost): - **Retain** GO:0004040 (amidase activity, MF, IEA/GO_REF:0000120) at the general level. - **Do not narrow** to acetamidase or any specific amide-hydrolase child term. - **Do not remove** the term — domain + structural homology is strong; ORF is "Verified." - Add a **curator note**: "Fungal-type AS amidase (PIRSF001221); intact catalytic triad (K176/S210/S238, AlphaFold pLDDT 96); substrate experimentally undetermined; closest characterized homolog = fungal acetamidase AmdS at 33% identity; distinct from canonical fungal AMI1 (absent from S288C)." - Verify GO_REF:0000120 is the correct current source vs EC2GO (GO_REF:0000003).

For experimentalists (to resolve the gaps): - Recombinant expression + amidase activity assay across a substrate panel (test #1). - *amd2Δ* nitrogen-source growth screen (test #2). - Triad mutagenesis to confirm catalytic mechanism (test #3).

Candidate references to verify (leads): - Founding reference

P 2263500

— confirm it is sequence-identification-only, to justify the IEA basis. - **PMID: 17924455** — verify the exact snippets *"conserved among plants, Bacillus subtilis, Neurospora crassa, Schizosaccharomyces pombe and Saccharomyces species, with the exception of S. cerevisiae S288C"* and *"suggesting that Ami1p may hydrolyse some amides related to amino acid and niacin metabolism in the cell."* - **PMID: 32024536** — supports plausibility of yeast acetamidase-type function; note it tests *Y. lipolytica*, not AMD2.

Bottom Line

AMD2 (P22580 / YDR242W) is, by strong and consistent domain/structural homology, a **fungal-type amidase-signature enzyme with an intact catalytic triad** — so the IEA annotation **GO:0004040 (amidase activity) is justified and should be retained at the general level**. But the assignment is **homology-only** (UniProt PE 3, SGD "Putative amidase," no assay), and the nearest characterized homolog (AmdS acetamidase, EC 3.5.1.4) is only 33% identical — too distant to assign a specific substrate. The correct curation action is **retain, keep general, flag as homology-based, substrate experimentally undetermined** — neither narrowing to acetamidase nor removing the term.
